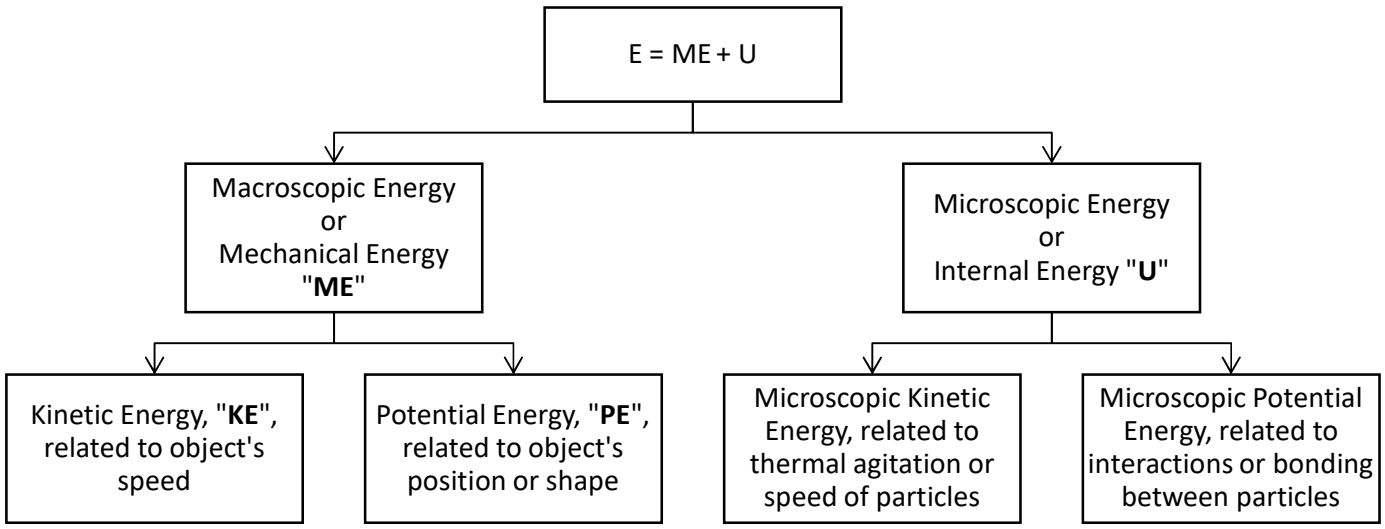
Chapter 1: Energy

Total Energy of a Mechanical System



As result, total energy: E = ME + U = KE + PE + U

Energy-Isolated System

A system is energy-isolated if its **total energy "E" remains constant over time**.

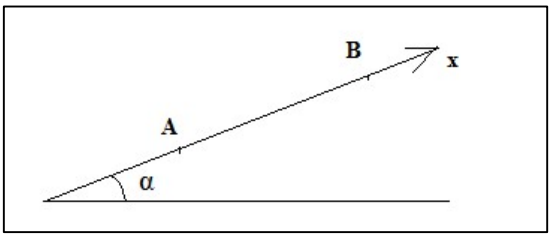
Since: E_1 (at t_1) = E_2 (at t_2), then $ME_1 + U_1 = ME_2 + U_2$, or $U_1 - U_2 = ME_2 - ME_1$, or $\Delta ME = -\Delta U$

- In addition, if friction is absent and there is no internal chemical reaction then U = constant $\Rightarrow U_1 = U_2$ $\Rightarrow U_2 - U_1 = 0 \Rightarrow \Delta U = 0 \Rightarrow \Delta ME = 0 \Rightarrow ME_2 - ME_1 = 0 \Rightarrow \boxed{ME_2 = ME_1 \text{ or } ME = \text{constant}}$; we say that mechanical energy is conserved.
- However, if friction exists ($\vec{f} \neq \vec{0}$) but there is no chemical reactions then $\Delta ME < 0$ (ME decreases over time and movement). As result: $\boxed{\Delta ME = W_{\vec{f}} = -\Delta U}$
- If both friction and chemical reactions exist, then: $\boxed{\Delta ME = W_{\vec{f}} + W_{\vec{F}}}$

  ($\vec{F}$ *refers to the traction or propulsion force that result from the chemical reaction*)

Application 1

An automobile climbs a slope of inclination α relative to the horizontal. Using the work-kinetic energy theorem, show that $\Delta ME = W_{\vec{f}} + W_{\vec{F}}$



Solution

The work-kinetic energy theorem applied between A and B, gives:

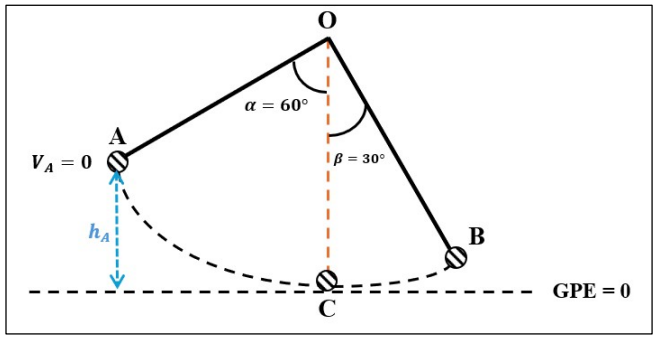
$\Delta KE = \sum W_{\vec{f}_{ext}} \Rightarrow \Delta KE = KE_B - KE_A = W_{\vec{w}} + W_{\vec{N}} + W_{\vec{f}} + W_{\vec{F}}$

But we know that: $W_{\vec{w}} = -\Delta GPE$, so: $\Delta KE = -\Delta GPE + W_{\vec{N}} + W_{\vec{f}} + W_{\vec{F}}$, or $\Delta KE + \Delta GPE = \Delta ME = W_{\vec{f}} + W_{\vec{F}}$

Application 2 (Simple Pendulum)

m = 20 g ; L = 80 cm ; $V_A = 0$

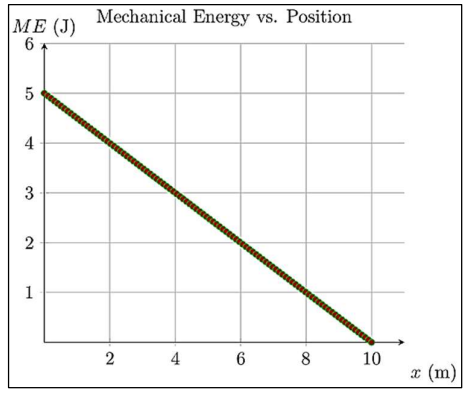
a) Find GPE(A) in terms of m, g, L, and α.
b) Calculate GPE(A).
c) Calculate GPE(B).
d) Deduce the variation of GPE between A and B.
e) Apply the law of conservation of ME to determine V_B.



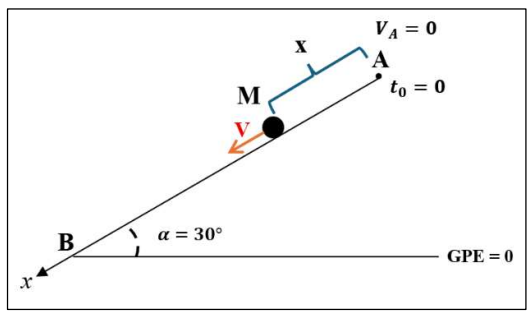
Application 3 (ME Graph)

An object of mass m = 100 g slides down an inclined plane making an angle α with the horizontal. A setup allows to register the variation of ME(J) as function of distance traveled x(m). The object was released without initial velocity from the origin of abscissa O.

a) Show that friction exists.
b) Find the expression of ME in term of x.
c) Find the magnitude f of the frictional force.



Application 4

Friction: f = 6 N; m = 2kg ; AB = 2 m

a) Determine V_B.
b) Let M the position of the mobile at any moment t, where its speed is V and its abscissa is x. Express the ME(t) of the system (mobile – earth) in term of m, g, v, x, and α.
c) Give the expression of ME(t) in terms of f and x.
d) Deduce the value of the acceleration.